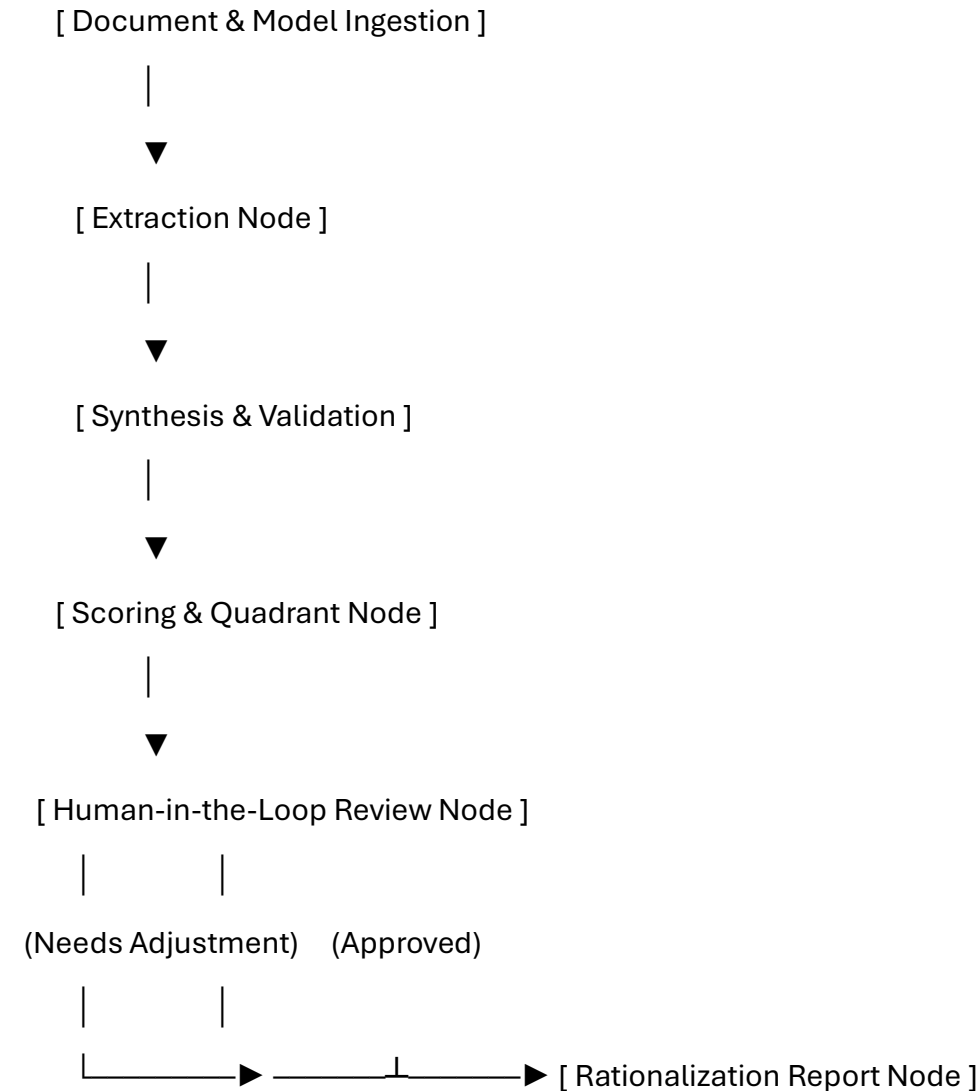


Application Portfolio Rationalization (TIME Matrix Engine)

1. System Architecture & Workflow (LangGraph Engine)

The engine uses **LangGraph** to model the Application Portfolio Rationalization process as a stateful, cyclic workflow. LangGraph handles multi-document contextual extraction, score calculation, rule-based validation, and human-in-the-loop auditability.



Graph Nodes Description

- 1. Ingestion & Parsing Node:** Receives uploaded documents (PDFs, Confluence exports, financial spreadsheets) and ArchiMate XML/JSON models. Extracts raw text, metadata, and structural relationships (e.g., application-to-business-capability mappings).

2. **Feature Extraction Node:** Uses LLM agents to extract target attributes per application:
 - **Technical Fit / Health:** Code quality, stack age, vendor support status, security vulnerability count, technical debt.
 - **Business Value:** Strategic alignment, user adoption, criticality, capability coverage.
 - **Run Cost:** License fees, hosting costs, operational maintenance burden.
 - **Risk Profile:** Compliance compliance status, single-point-of-failure status, SLA breaches.
3. **Score Calculation & Mapping Node:**
 - Aggregates raw attributes into two core composite metrics: **Technical Health (\$X\$-axis)** and **Business Value (\$Y\$-axis)**.
 - Maps coordinates to the corresponding TIME quadrant:
 - High Technical Health + High Business Value $\rightarrow$ **Invest**
 - Low Technical Health + High Business Value $\rightarrow$ **Migrate** (Modernize/Re-platform)
 - High Technical Health + Low Business Value $\rightarrow$ **Tolerate**
 - Low Technical Health + Low Business Value $\rightarrow$ **Eliminate**
4. **Human-in-the-Loop Node:** Suspends graph execution when confidence scores fall below threshold or when high-cost decisions (Eliminate/Migrate) are proposed. Accepts explicit user input to adjust placement or override metrics.
5. **Report Generation Node:** Produces consolidated executive summaries, overlap analyses, and action roadmaps.

2. Document & ArchiMate Context Binding

To yield defensible ratings, every score generated by the engine requires direct traceability back to source artifacts.

Source Type	Ingestion Strategy	Extracted Context
ArchiMate Models	Parse XML/Open Exchange format into a graph structure	Application Functions, Realized Business Capabilities, Interfaces, Infrastructure Services
Vendor Contracts & Financials	Vector Embeddings + Structured extraction (RAG)	Annual licensing cost, contract renewal dates, SLA clauses
Technical & Architecture Docs	Chunked RAG + Structural Chunking	Tech stack versions, integration patterns, security vulnerabilities
Business Operations Audits	Semantic Search	Active user count, process criticality, redundant capability tags

3. User Interface (UI) Specification

The interface consists of a 3-panel workspace: **Navigation & Catalog**, **Interactive TIME Matrix**, and **Contextual Evidence Drawer**.

Layout Structure

```
+-----+
| [Header] App Portfolio Rationalization (TIME Engine) [Filter: Business Unit v] [Export v] |
+-----+-----+-----+
| Catalog Filter & List | Main Panel: Interactive Matrix | Context & Evidence Panel | | | |
|           |           |           |
| Search: [_____] | Business Value ^ | App: Core CRM Platform |
|           | (10)           | Quadrant: INVEST |
| [ ] ERP Core | +-----+-----+ |           |
| [x] Core CRM | | MIGRATE |INVEST| | **Scores & Metrics** |
```


- Provides a confidence indicator (e.g., *High confidence: 3 matching documents* vs. *Low confidence: Single outdated document*).

3. Duplicate Capability Detector

- Highlights applications occupying identical business capability nodes within linked ArchiMate models.
- Generates side-by-side comparison tables highlighting redundant licensing costs and recommending consolidation targets.

4. Scoring Methodology & Decision Rules

To ensure predictable outputs, scoring relies on weighted composite index calculations combined with deterministic threshold logic:

$$\text{Technical Health} = w_1 \cdot \text{Architecture Modernity} + w_2 \cdot \text{Vendor Supportability} - w_3 \cdot \text{Vulnerabilities} - w_4 \cdot \text{Maintenance Effort}$$

$$\text{Business Value} = v_1 \cdot \text{Capability Coverage} + v_2 \cdot \text{User Impact} + v_3 \cdot \text{Strategic Alignment} - v_4 \cdot \text{Redundancy Index}$$

Quadrant Rules

Quadrant	Score Thresholds	Action Trigger
Invest	Tech ≥ 5.0 , Value ≥ 5.0	Maintain funding; expand coverage to replace 'Migrate' apps.
Tolerate	Tech ≥ 5.0 , Value < 5.0	Freeze development; reduce operational costs; re-evaluate annually.
Migrate	Tech < 5.0 , Value ≥ 5.0	Refactor, re-platform, or replace tech stack while maintaining business logic.
Eliminate	Tech < 5.0 , Value < 5.0	Phase out, decommission, and reassign remaining capabilities to 'Invest' systems.